

Global Pangenome Analysis Highlights the Critical Role of Structural Variants in Cattle Improvement and Identifies a Unique Event as a Novel Enhancer in *IGFBP7*⁺ Cells

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Abstract

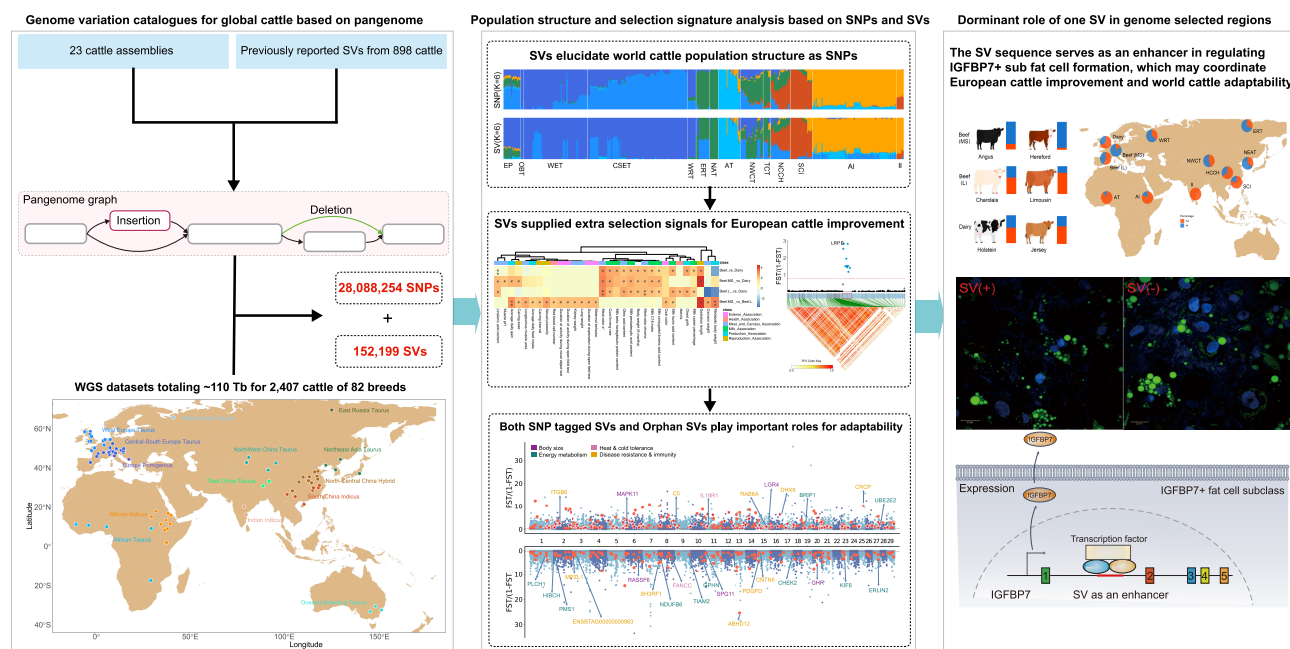
Based on a pangenome graph platform, we simultaneously analyzed the impacts of SNPs and SVs in the population structure and phenotypic formation of global cattle using 2,409 individuals from 82 breeds. We demonstrated that SVs, like SNPs, effectively explain the population structure of global cattle. Genomic regions under strong selection, identified using both SNPs and SVs, consistently revealed footprints associated with human-mediated selection of economic traits in European improved cattle or natural selection of geographical adaptations. Notably, we detected that ~40.14% of SVs were not tagged ($LD, r^2 < 0.6$) by nearby SNPs. These “orphan” SVs may uncover new genetic signals and represent recent mutations associated with specific selection pressures or local environmental adaptation. Selected SVs tagged by SNPs also play causal or dominant roles in regions under selection. For example, our single-cell RNA sequencing has demonstrated that a notable SNP-tagged SV functions as an enhancer of the *IGFBP7* gene, regulating fat deposition through IGFBP7⁺ cells. In conclusion, these SV-related mechanisms likely have caused some differences in economic traits and local adaptability across global cattle populations. Our integrated approaches highlight the unique and indispensable roles of SVs in shaping genetic diversity, offering novel insights into adaptation, selection, and strategies for improving cattle populations.

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Graphical Abstract



Keywords: cattle, pangenome, structural variation (SV), population differentiation, single cell sequencing, *IGFBP7*

Introduction

Cattle are among the most economically and culturally significant domesticated species, contributing to food security, agriculture, and livelihoods worldwide. Due to natural and human selection, the global cattle populations have shown abundant phenotypic diversity. These phenotypic differences related to traits, especially adaptability and economic traits, contribute to differences in the economic value of cattle (Hayes and Daetwyler 2019). Generally, modern cattle trace their origins to multiple domestication events from wild aurochs populations in distinct geographic regions (Rossi et al. 2024). Two primary domestication events shaped the species: one approximately 10,000 years ago in the Fertile Crescent, giving rise to humpless taurine cattle (*Bos taurus* or *B. taurus*), and another around 8,000 years ago in the Indus Valley, resulting in humped indicine cattle (*Bos indicus*, *B. indicus* or Zebu cattle) (Pitt et al. 2019). Taurine and indicine cattle are interfertile and have experienced historical and recent hybridization. Indicine cattle have unique advantages in adapting to hot environments that have been facilitated by humans in response to a multi-century drought approximately 4,200 years ago (Verdugo et al. 2019). In Europe, taurine cattle underwent continuous breeding for economic gains such as milk and meat production after the Agricultural Revolution, and achieved great success, developing multiple world-renowned breeds. Compared to those in Europe, most of the cattle in Asia and Africa have not undergone targeted breeding due to conscription and nonintensive feeding.

These divergent traits are likely driven by subspecies-specific alleles or shared variations in various allele frequencies, which underlie unique quantitative trait loci

(QTL) associated with these characteristics (Bolormaa et al. 2013). A large number of studies focused on the diversity of cattle populations and the uneven frequency distribution through genomics analyzing, which would draw wide interest and benefit the genetic improvement of the cattle industry (Karim et al. 2011; Chen et al. 2018a; Hayes and Daetwyler 2019; Kim et al. 2020). However, at least two factors limit our study from getting more comprehensive and accurate results. The first one is the limitation of the nucleotide sequence of the reference genome based on a Hereford cow. Even though it has undergone multiple upgrades since its initial release (Consortium 2009; Zimin et al. 2009; Rosen et al. 2020), the single reference genome does not fully capture cattle genetic diversity. For example, it is not yet telomere-to-telomere quality, while the best efforts so far are the genome from the Wagyu breed, which now has complete T2T sex chromosomes and a few autosomes (<https://www.researchsquare.com/article/rs-6068440/v1>). The second idea that has been proposed is the pangenome, which incorporates broader genetic diversity beyond a single reference genome (Smith et al. 2023). In cattle, a previous study developed breed-specific and pangenome reference graphs, uncovering 70 Mb of novel sequences (Crysnanto et al. 2019; Crysnanto and Pausch 2020; Crysnanto et al. 2021; Gao et al. 2025). Another study added 116 Mb of African cattle sequences, enhancing read mapping and variant calling accuracy (Talenti et al. 2022). Leonard et al. reported consistent pangenomes across platforms and constructed a pangenome from 16 PacBio HiFi assemblies (Leonard et al. 2022, 2023).

A pangenome is defined as the nonredundant collection of all DNA sequences present in a population, including a

“core” genome shared by all individuals and a “variable” genome found only in a subset (Tettelin et al. 2005). The “variable” genome, including structural variation (SV), significantly contributes to overall genetic divergence (Eichler et al. 2007). SV involves larger genomic segments compared to single nucleotide polymorphisms (SNPs) and short insertion/deletion (InDel) variants. It includes deletions, insertions, duplications (commonly grouped as copy number variations or CNVs), inversions, and translocations, or complex events (involving two or more different types), ranging from 50 base pairs (bp) to 5 megabase pairs (Mb) (Scherer et al. 2007). The large size of SV increases the likelihood of influencing gene expression and function by altering gene dosage, interrupting coding sequences, or disturbing long-range regulation (Alkan et al. 2011; Sudmant et al. 2015). Previous research has detected SV in cattle genomes and linked them to potential phenotypes (Liu et al. 2010; Bickhart et al. 2012; Cicconardi et al. 2013; Kommadath et al. 2019; Chen et al. 2020; Hu et al. 2020; Low et al. 2020; Jang et al. 2021; Lee et al. 2021; Upadhyay et al. 2021). However, most of the previous research was mainly conducted on SNPs, the research on SV has just become popular recently in the era of pangenome research (Zhou et al. 2022; Leonard et al. 2024). It has been proven that more than 24% of SVs showed no or low linkage disequilibrium (LD) with SNPs in humans, and the SV could independently play important roles in genome biology and complex traits (Pang et al. 2010). Thus, one current limitation is that other kinds of mutations, especially SV, have not been fully studied, which results in unclear understanding of the diverse phenotypic differences in cattle at present.

In this study, we integrated SNP and SV to provide a comprehensive understanding of the genetic architecture of global cattle populations based on a pangenome graph platform. Using short-read whole-genome sequencing data from 2,409 individuals spanning 82 breeds, we investigated population structure, selective pressures, and adaptive traits. We identified genomic regions under selection associated with economically important traits and environmental adaptation. We examined both selected SVs, whether tagged or untagged by SNPs, for their potential functional roles and as candidates for recent favorable mutations. Finally, we validated the enhancer activity of one SV, which might regulate the cell-type-specific expression of *IGFBP7* using single-cell RNA sequencing. Overall, our study underscores the complementary roles of SNPs and SVs in shaping genetic diversity and offers valuable insights into adaptation, selection, and strategies for livestock improvement, particularly in developing regions where environmental challenges are pronounced.

Results

Phenotype and Genome Variations of the Collected Datasets From Worldwide Cattle Breeds

We collected WGS datasets totaling ~110 Tb for 2,409 cattle of 82 breeds, which represented the extensive genetic diversity of global cattle populations (Fig. 1a and supplementary table S1, Supplementary Material online). We classified the cattle into 13 different populations according to their geographic origins and taxonomies. Additionally, we retrieved reliable phenotype information for each cattle breed through trustworthy sources and publications (supplementary table S2, Supplementary Material online). These cattle exhibited substantial differences in economic phenotype and adaptability

due to their diverse origins and levels of selective breeding. These include several widely hailed economic body traits of cattle, such as milk production, body weight, body height, and their adaptation to different climates (Fig. 1a and supplementary table S2, Supplementary Material online). Notably, cattle in Asia and Africa showed significantly lower beef and milk production compared to European cattle, which have undergone hundreds of years of breeding and selection (Fig. 1a and supplementary fig. S1a, and b, Supplementary Material online). However, European cattle also face challenges, such as adapting to high-temperature climates, when they are introduced into southern Asia and Africa (Fig. 1a and supplementary fig. S1c, Supplementary Material online).

While the 1,000 Bulls Project had a slightly higher total sample count (Hayes and Daetwyler 2019), our study emphasizes diversity across major global lineages and complements that resource by including populations that were underrepresented in previous datasets. We analyzed two types of important genomic variations (SNPs and SVs) to comprehensively study the genetic diversity corresponding to worldwide cattle. Using a standardized SNP calling pipeline via GATK (McKenna et al. 2010), we identified and genotyped 28,088,254 SNPs with minor allele frequency (MAF) ≥ 0.01 across all samples using ARS-UCD1.2 as the reference. For SVs, we built a cattle pangenome graph by integrating 23 cattle assemblies and 92,518 previously reported SVs (supplementary table S3, Supplementary Material online). In total, 152,199 SV events were genotyped as biallelic variations (0/0, 0/1, and 1/1) for each animal using the VG software with the pangenome graph as the reference (Hickey et al. 2020). Compared to the previous study (Zhou et al. 2022), we identified 75,774 novel SVs, including 32,769 deletions (absences) and 43,005 insertions (presences). The inserted sequences had a total length of 24.51 Mb, which was absent in the reference genome. Over 82.61% (76,425) of the deletions previously reported (Zhou et al. 2022) were re-detected in this study. The call rate improved by an average of 14.26% using the VG software compared to our previous strategy (Zhou et al. 2022) (supplementary fig. S2a and S2b, Supplementary Material online). The SV genotyping results showed high consistency (93.38%) with our previous method (supplementary fig. S2c, Supplementary Material online). Moreover, 93.75% of the homozygous mutants and 83.75% of the heterozygous mutants were confirmed through targeted PCRs (supplementary table S4, Supplementary Material online). Generally, 139,036 (91.35%) SVs were supported by at least two animals. Considering a one-Mb distance, each SV was surrounded by at least 7,085 SNPs, with an average of 22,602 SNPs used for further analysis (supplementary fig. S2d, Supplementary Material online). This dataset provides a high-quality resource for comprehensive studies on the genetic diversity of cattle populations based on both SNPs and SVs.

SVs Reveal Global Cattle Population Structure Comparable to SNPs

To elucidate relationships among cattle populations, we conducted PCA, clustering, admixture, and gene drift analyses. Both SNP-based and SV-based population taxonomy analyses yielded consistent results. In the PCA analysis, the first component PC1 divided cattle into taurine and indicine, with hybrids positioned between them (Fig. 1b and supplementary fig. S3a, Supplementary Material online). The second component clearly separated the cattle from India, Africa, and South

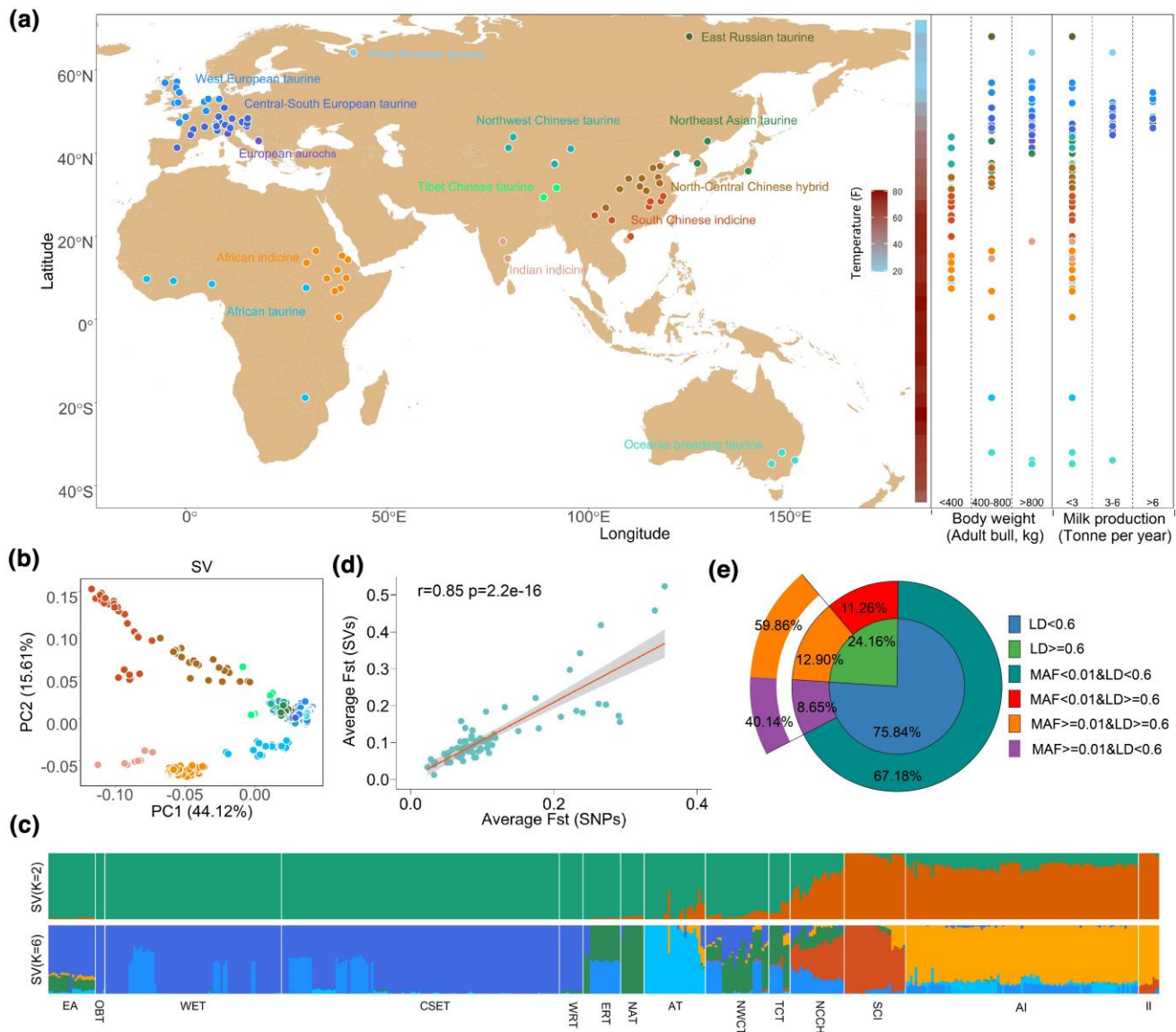


Fig. 1. Population structure and genetic diversity of the worldwide cattle breeds. a) Geographic classification, distribution, and phenotypes of the cattle breeds analyzed in this study. b) PCA analysis based on SVs. c) Admixture analysis based on SVs. EA, European aurochs; OBT, Oceania breeding taurine; WET, West European taurine; CSET, Central-South European taurine; WRT, West Russian taurine; ERT, East Russian taurine; NAT, Northeast Asian taurine; AT, African taurines; NWCT, Northwest Chinese taurines; TCT, Tibet Chinese taurine; NCCH, North-Central Chinese hybrid; SCI, South Chinese indicine; AI, African indicine; II, Indian indicine. d) Correlation analysis of average F_{ST} values across 28 different cattle population comparisons based on SNPs and SVs. e) Pie chart showing the percentage of SVs tagged by SNPs (r^2 for linkage disequilibrium ≥ 0.6 , within 1 Mb of the SV).

China within indicine and the cattle from Europe and West Africa within taurine (Fig. 1b and supplementary fig. S3a, Supplementary Material online). Clustering analysis supported these PCA findings and aligned with the historical distribution of cattle (supplementary fig. S3b, Supplementary Material online). For example, Australian Lowline cattle were derived from European Angus cattle, while West Russian cattle were similar to West European taurine, and the East Russian cattle shared similarities with Northeast Asian taurine (Fig. 1b and supplementary fig. S3a and b, Supplementary Material online).

In the admixture analysis, at $K=2$, taurine and indicine cattle were successfully distinguished by both SNPs and SVs (Fig. 1d and supplementary fig. S3c, Supplementary Material online). At $K=6$, although SVs did not separate West Europe and Central-South Europe taurine as SNPs did, they accurately explained the relationship of other cattle breeds, identifying taurine in Europe, Northeast Asia, and Africa, as

well as indicine in North China and Africa (Fig. 1d and supplementary fig. S3c, Supplementary Material online). Chinese cattle showed the most complex ancestry. Northwest Chinese taurine showed mixed contributions from European taurine and Northeast Asian taurine, while North-Central China hybrids combined ancestry from European taurine, Northeast Asian taurine, and South Chinese indicine. In Africa, as previously reported, taurine and indicine cattle were partially crossed with each other. Both the gene drift analyses by SNPs and SVs showed that the Chinese and African cattle were both affected by European cattle breeds, alongside regional gene flow within their respective areas (supplementary fig. S3d, Supplementary Material online).

We calculated the average F_{ST} values for 74 population pairs, finding a strong correlation (0.85) between SNP- and SV-based results (Fig. 1d and supplementary table S5, Supplementary Material online). However, when considering

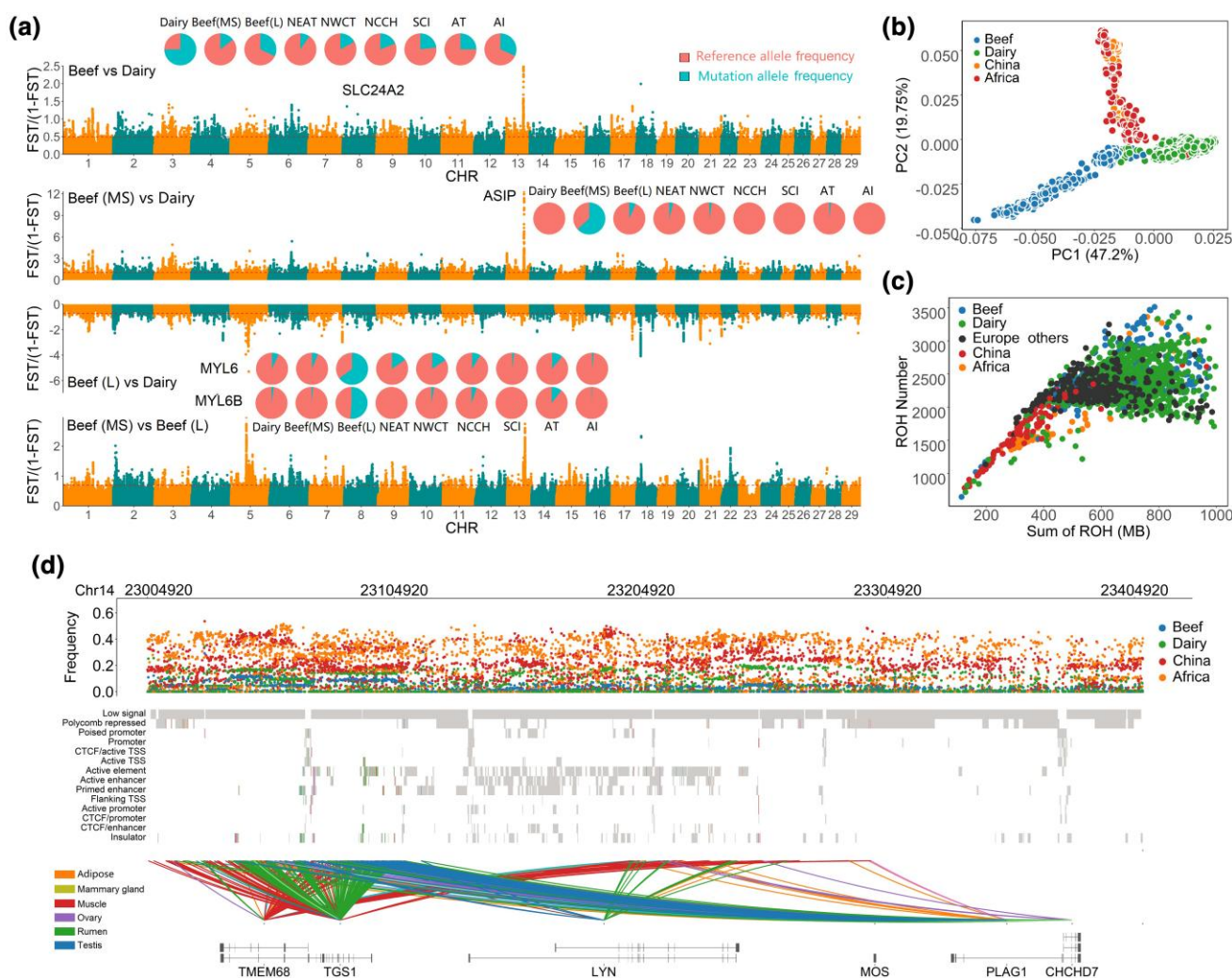


Fig. 2. Selection signatures of SNPs in European beef and dairy cattle. a) Manhattan plot of F_{ST} values from SNP-based selection scans across four comparisons: beef cattle vs. dairy cattle, medium-small (MS) beef cattle vs. dairy cattle, large (L) beef cattle vs. dairy cattle, medium-small (MS) beef cattle vs. large (L) beef cattle. Dairy: European improved dairy cattle; Beef (MS): European improved medium-small beef cattle; Beef (L): European improved large beef cattle; NAT, Northeast Asian taurine; NWCT, Northwest Chinese taurine; NCCH, North-Central Chinese hybrid; SCI, South Chinese indicine; AT, African taurine; AI, African indicine. The dotted line represents the threshold for the top 1%. b) PCA analysis using the top 1% F_{ST} SNPs between beef cattle and dairy cattle; Beef: European improved beef cattle; Dairy: European improved dairy cattle; Europe others: cattle breeds except the improved beef and dairy cattle; China: cattle breeds in China; Africa: cattle breeds in Africa. c) Dot plot showing ROH length and count across different cattle populations; Beef: European improved beef cattle; Dairy: European improved dairy cattle; China: cattle breeds in China; Africa: cattle breeds in Africa. d) SNP heterozygous frequency and potential regulation mechanisms integrating chromatin states and FarmGTEX EWAS data for a beef ROH hotspot (chr14:23004920–23408215). CattleGTEX and FAANG results, including eSNPs, eGenes, and chromatin states, were graphically visualized to explore potential enhancer signals. While eSNP–eGene relationships were found to be widespread due to their complex, many-to-one and one-to-many nature, the pairings were clearly defined. Muscle, rumen, and testis tissues from beef cattle, as well as adipose and rumen tissues, were included in the analysis, with tissue-specific pairings indicated by line colors.

the global LD (measured as r^2) between SNPs and SVs, we found a large portion (75.84%) of SVs could not be tagged by SNPs, most of which were with low MAF < 0.01 . Only 59.86% of SVs with MAF ≥ 0.01 were tagged ($r^2 \geq 0.6$) by at least one SNP within a one-Mb distance around SVs (Fig. 1e). Overall, besides their contributions captured by SNPs, our results also emphasize the independent and significant role of SVs in global cattle population classification and their potential influence on phenotypic variation.

SNP-based Selection Signals in European Cattle Guide Genomic Improvement in Unselected Populations

European improved cattle breeds have undergone systematic, targeted breeding for beef and milk production during the past

centuries, leaving genome footprints of selection that are valuable for directing cattle breeding in other regions. These improved cattle breeds exhibited lower decay of LD for both SNPs and SVs and had larger regions of homozygosity (ROH) compared to other European cattle breeds (supplementary fig. S4, Supplementary Material online and Fig. 2c). We grouped animals by breeding purpose and size: dairy cattle (e.g. Holstein, Jersey, German Black Pied, Brown Swiss, Fleischvieh, Gelbvieh, Rendena, Normande, totally 1,341 individuals), medium-small (MS) beef cattle (e.g. Angus, Hereford, totally 201 individuals), and large (L) beef cattle (Charolais, Limousin, Belgian Blue, Piemontese, totally 158 individuals). We applied a combined strategy of F_{ST} , XP-CLR, and XP-EHH based on SNPs to detect the selected genomic regions among these three populations (Fig. 2a). In total, 223, 237, 223, and 273 nonoverlapping regions were

identified using 22.85 Mb SNPs between Beef and Dairy cattle, Beef (MS) and Dairy cattle, Beef (L) and Dairy cattle, and Beef (MS) and Beef (L) cattle, respectively (supplementary table S6, Supplementary Material online).

We examined the potential effects of selective signatures by comparing their chromosomal positions to cattle QTLs found in the public database that includes 192,247 QTLs for 552 different base traits from 1,179 publications (Hu et al. 2021). As expected, the significantly selected windows were enriched with milk content-related traits (milk beta-casein percentage, milk lactose content, etc.) between dairy cattle and all beef cattle, and with meat and carcass-associated traits (average daily gain, subcutaneous fat, carcass weight, etc.) between the two kinds of beef cattle (supplementary fig. S5, Supplementary Material online). Functional annotation of genes within these selected regions showed that they were partially enriched in the GO terms or pathways related to specific characteristics of two populations compared (supplementary table S7, Supplementary Material online). This included protein binding, calcium ion import across the plasma membrane, etc. between beef cattle and dairy cattle, and ATP binding, lipid binding, vascular smooth muscle contraction, thyroid hormone synthesis, etc. between MS and L beef cattle.

Interestingly, when using the top 1% SNPs derived from the F_{ST} analysis between European beef and dairy cattle to perform PCA analysis, we observed both Chinese and African cattle positioned between European beef and dairy cattle along PC1 (Fig. 2b). This supports the notion that Chinese cattle and African cattle have not undergone targeted breeding for dairy and meat production, i.e. the important genomic variations underlying beef and milk production in Chinese and African cattle were not under selection pressure. For example, the SNPs, located in *SLC24A2* (chr8:24555356) selected for calcium channel activity in dairy cattle, *ASIP* (chr13:63667387) selected for fat deposition in beef (MS) cattle, and *MYL6* (chr5:57162869) and *MYL6B* (chr5:57161801) selected for muscle development in beef (L) cattle (Magalhães et al. 2016; Liu et al. 2019; da Cruz et al. 2020), were all in the opposite ends of frequency distributions to the selected European cattle populations (Fig. 2a). This implied that the candidate selection signatures for Europe's improved beef and dairy cattle would be useful to guide the target breeding for Chinese and African cattle.

European improved cattle had significantly longer and more ROH regions compared to Chinese and African cattle (Fig. 2c). We identified 45 and 36 ROH hotspots (shared in over 80% of cattle and ≥ 100 kb) for European improved beef and dairy cattle, respectively, in which the Chinese and African cattle both showed low ROH frequency (supplementary tables S8 and S9, Supplementary Material online). These ROH hotspot regions were enriched for beef and milk production-related QTLs and genes. We identified an ROH hotspot specific for beef cattle in chr14:23004920–23408215. It covered multiple QTLs for carcass weight, average daily gain, body weight, fat thickness at the 12th rib, etc., as well as six genes including *PLAG1*, *CHCHD7*, *MOS*, *LYN*, *TGS1*, and *TMEM68* (Fig. 2d and supplementary table S10, Supplementary Material online). Multiple studies have proved that this region, including *PLAG1* is strongly related to the growth and body size of cattle (Karim et al. 2011; Nishimura et al. 2012; Engle and Hayes 2022). In dairy cattle, we also detected a specific ROH hotspot in chr18:56932170–57022131 covering dozens of the *KLK* gene family members (supplementary fig. S6, Supplementary Material online and

supplementary table S10, Supplementary Material online). Even though *KLK* genes were not reported to have a direct association with milk production, they were proven to play key roles in the health of the mammary gland (Pampalakis et al. 2019; Christiono and Anggarani 2021). Cross-referencing with the 13 chromatin states and eQTLs from FarmGTEx, we successfully located the potential causal mutations and regulator elements for the ROH hotspots-covered genes. Notably, eQTL signals were especially enriched in several tissues (muscle, rumen, and testis for beef cattle, adipose, and rumen for dairy cattle) corresponding to the improvement of beef and dairy production (Fig. 2d and supplementary fig. S6, Supplementary Material online, supplementary table S10, Supplementary Material online). For example, enhancers located in chr14:23092257–23095329 might regulate the expression of *PLAG1* in the testis which might be the actual causal region for body weight in cattle (Reverter et al. 2020; Engle and Hayes 2022) (Fig. 2d).

Recent SV Mutations Provide Genetic Variations Beyond SNPs for the Improvement of European Beef and Dairy Cattle

We detected 2,423 selective SV signatures from 116,194 SVs by applying F_{ST} statistics with a top 1% threshold across all comparisons among European improved beef and dairy cattle (supplementary table S11, Supplementary Material online). Notably, 32.80% to 60.12% of genes uncovered by SVs were not detected by SNPs using similar thresholds (top 1% F_{ST} value) (supplementary table S12, Supplementary Material online). At the same time, we note that a substantial portion of selective SNP signals ($\sim 95\%$) could not be tagged by SVs. For example, genes like *ASIP*, *MYL6*, and *MYL6B*, detected by SNPs, were not identified by SVs. Nevertheless, annotated genes affected by selected SV signatures were enriched in functions similar to those candidate SNPs (supplementary table S13, Supplementary Material online). Significantly selected SV regions were enriched with QTLs related to milk production and meat and carcass traits (Fig. 3a). This confirmed that the SVs are as significant as SNPs and may explain phenotypic variations in cattle that SNPs alone cannot account for.

Generally, selective SNP signals became stronger as the distance to selected SV signals decreased, consistent with selection sweeps (supplementary fig. S7, Supplementary Material online). However, not all selected SVs could be tagged by nearby SNPs. We defined the SVs without tagged SNPs ($r^2 < 0.6$, distance ≤ 1 Mb) as orphan SVs. We identified on average 1,395 (57.57%) orphan SVs and 1,082 (42.47%) SNP-tagged SVs across the four comparisons. Over half (55.30%) of the selected SVs overlapped with MEIs (mobile element insertions) in the genome. A detailed analysis of MEI types revealed that bovine-specific MEIs, including L1_BT (43.51%), BovB (13.25%), and BovA2 (10.32%), contributed most to the selective SV signatures (supplementary fig. S8, Supplementary Material online and Fig. 3b). L1_BT was particularly prominent among orphan SVs, exhibiting significantly younger divergence compared to its SNP-tagged counterparts (Fig. 3c), suggesting that orphan SVs represent more recent i.e. younger events.

Among the selected SV signatures distinguishing dairy and all beef cattle, we observed a top selected orphan SV (chr2:56290045–56298615) located in the intron of the *LRP1B* gene (Fig. 3d and e). *LRP1B* (low-density lipoprotein-related

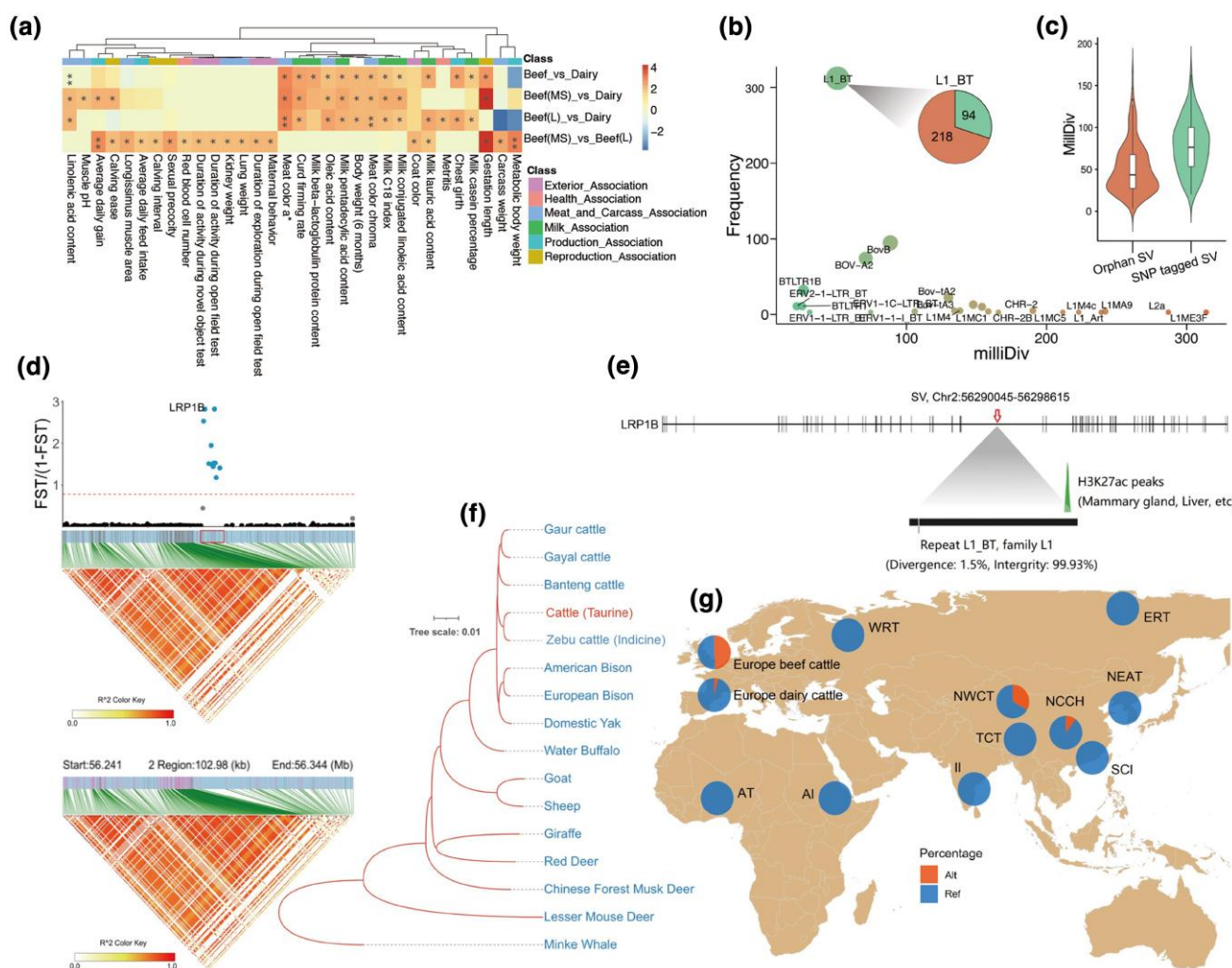


Fig. 3. SV contributions to the genetic improvement of European beef and dairy cattle. a) Heatmap of QTL enrichment in significantly selected SV regions. b) Frequency and divergence of different MEI elements among significantly selected SVs. c) Box plot comparing the divergence of L1_BT between orphan SVs and SNP-tagged SVs. d) Partial Manhattan plot showing significantly selected orphan SVs in *LRP1B* between all beef cattle and dairy cattle. The LD block is plotted with or without significant SVs. e) Genomic position and sequence characteristics for the SV in the *LRP1B* gene. f) Manual inspection of genome assemblies across ruminants for the existence of the L1_BT sequence within the orphan SV. The species name with red color represented L1_BT presence and the species name with blue color represented absence. g) Frequency distribution of the orphan SV across cattle populations with different geographic and taxonomic classifications. WRT, West Russian taurine; ERT, East Russian taurines; NAT, Northeast Asian taurine; NWCT, Northwest Chinese taurine; TCT, Tibet Chinese taurine; NCCH, North-Central Chinese hybrid; SCI, South Chinese indicine; II, Indian indicine; AT, African taurine; AI, African indicine.

protein 1B) has been implicated as a candidate gene for fat deposition in milk (Martins et al. 2021). This orphan SV consisted of a complete L1_BT sequence with low divergence (1.5%) and contained H3K27ac enhancer signals in tissues such as the mammary gland and liver (Fig. 3e). Manual inspection of genome assemblies across ruminants revealed that this SV insertion occurred exclusively in taurine cattle (Fig. 3f). Population variation analysis further suggested that this L1_BT insertion likely arose after the divergence of European taurine and other taurine populations (Fig. 3g). Admixture analysis indicated that the SV variation in Northwest Chinese taurine and North Central Chinese hybrid cattle likely resulted from European taurine introgression (Figs. 1d and 3g). Thus, we hypothesize that this orphan SV represents a recent L1_BT insertion in the European taurine population, introducing functional elements that modulate *LRP1B* expression, thereby influencing milk fat deposition. During subsequent selection processes, this mutation diverged between dairy and beef cattle.

SNP and SV Selection Signatures Jointly Explain the Challenges Faced by European Cattle When Introduced to Tropical Regions

The introduction of improved dairy and beef cattle from Europe to regions lacking high-performance breeds is often complicated by challenges in adapting to local environmental and climatic conditions. The primary challenge for European cattle is environmental adaptation, including heat and disease resistance, when they are imported to South China and Africa. We detected 2,322 SVs specific to indicine cattle in these regions (absent in taurine populations, frequency ≥ 0.1 in indicine populations), contributing 1.19 Mb of novel sequences and overlapping with 588 gene body regions (supplementary table S14, Supplementary Material online). By performing selective signature scanning, we detected 25,743 SNPs (in 890 selected 50 kb-windows) and 2,081 SVs (1,050 presence and 1,031 absence events) as significantly selected between European taurine and South Chinese/African indicine cattle

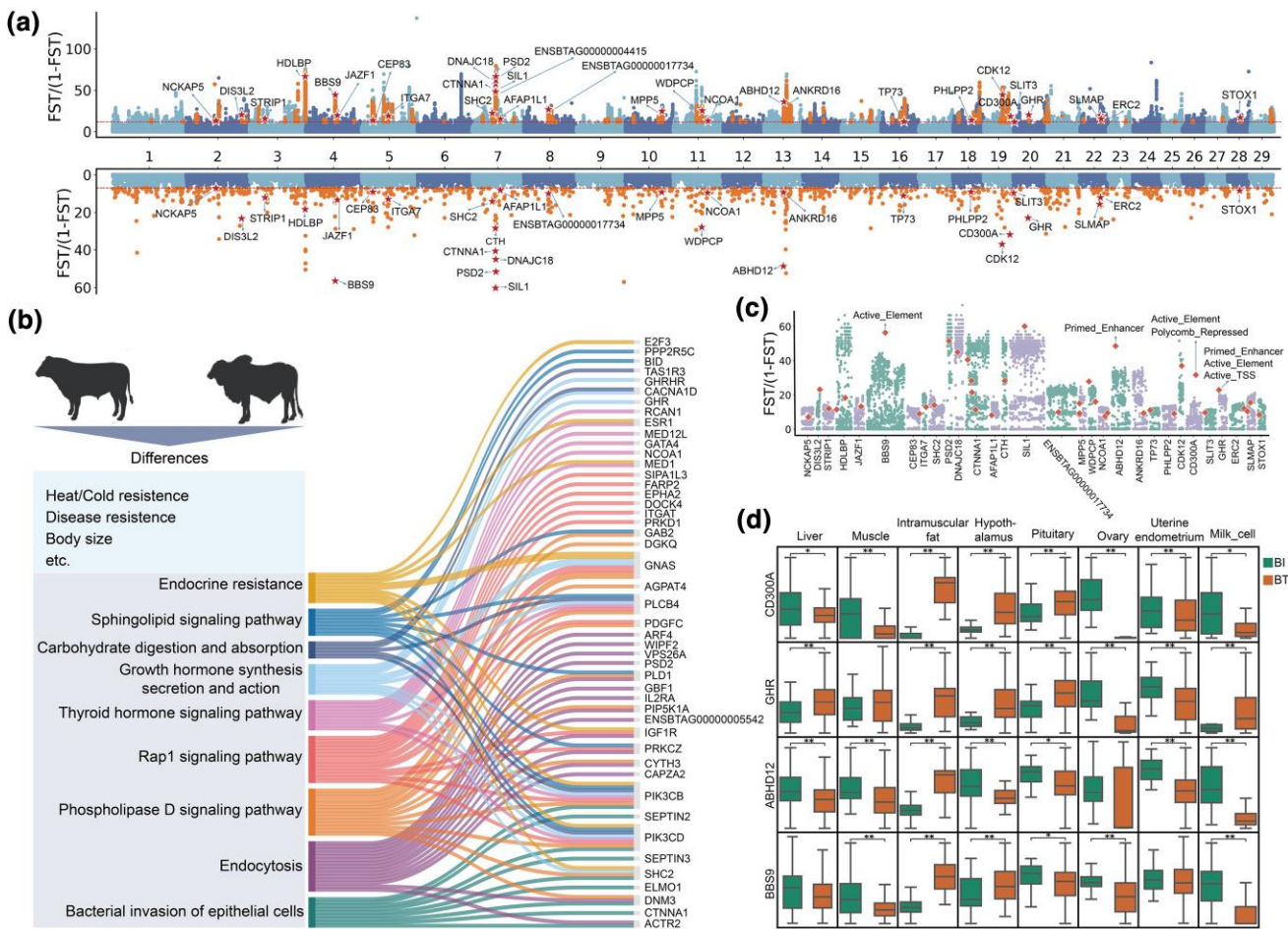


Fig. 4. Dominant effects of SVs in selected genome regions between European taurine and Chinese-African indicine. a) Manhattan plot displaying selection signatures of SNPs and SVs. Significantly selected variants are highlighted in orange, with commonly selected genes labeled with stars and the gene names. b) Phenotypic differences and functional pathways affected by significant SVs between European taurine and Chinese-African indicine. c) Distribution of F_{ST} values and overlap with functional genomic elements for selected SVs. d) Expression differences between taurine and indicine cattle in different tissues for the four genes with functional elements overlapping SVs. The student's t-test was used for the significance test. “*” $P < 0.05$; “***” $P < 0.01$.

populations. Almost all selected SVs between European taurine and South Chinese/African indicine were tagged by SNPs, and the defined orphan SVs have LD $r^2 < 0.6$ with SNPs (supplementary fig. S10a, Supplementary Material online). There were 30 genes commonly annotated by significantly selected SNPs and SVs (Fig. 4a). The previously identified selected genes, including *CTNNA1*, *DNAJC18*, etc. were both uncovered by SNPs and SVs simultaneously. Genes overlapped with significant SNPs and SVs were most enriched in Bacterial invasion of epithelial cells pathway and Endocytosis pathway for disease resistance; Thyroid hormone signaling pathway, Phospholipase D signaling pathway, Rap1 signaling pathway, Sphingolipid signaling pathway for heat/cold resistance; as well as Growth hormone synthesis, secretion and action pathway and Carbohydrate digestion and absorption pathways for body sizes (Fig. 4b and supplementary table S15, Supplementary Material online).

However, even if SVs and SNPs uncovered the same genes with selection pressure, SV might play key roles in those selection regions for differences between European taurine and indicine. We compared all F_{ST} values for SNPs and SVs that overlapped with genes. We saw SVs with the highest F_{ST} value in 7 of the 30 genes with selection pressure and 4 genes (*BBS9*, *ABHD12*, *CD300A*, and *GHR*) that directly changed the

sequence structure of functional elements including a primed enhancer, active element, active TSS and polycomb repressed (Fig. 4c). To figure out if the four selected SVs would affect the expression of the genes, we compared the gene expression levels of the four genes between 7,105 taurine and 490 indicine in eight different tissues from CattleGTEx. The results showed that all four genes were widely differentially expressed in different tissues between taurine and indicine cattle (Fig. 4d), implying that some SVs may play dominant roles in the genome regions with selective pressure. We acknowledge that future studies integrating tissue and breed matched epigenomic data will improve the precision of functional interpretation.

Novel SV Mutations Promote Subpopulation of Local Cattle Adaptability

We further performed selective signature scanning for both SNPs and SVs of different cattle populations according to different taxonomy relationships and geographical regions. Totally, we grouped the animals into 28 different pairs within 8 sub-groups (supplementary table S16, Supplementary Material online). We detected 3,301 regions (spanning 312.81 Mb in length) and 6,573 SVs that were significantly

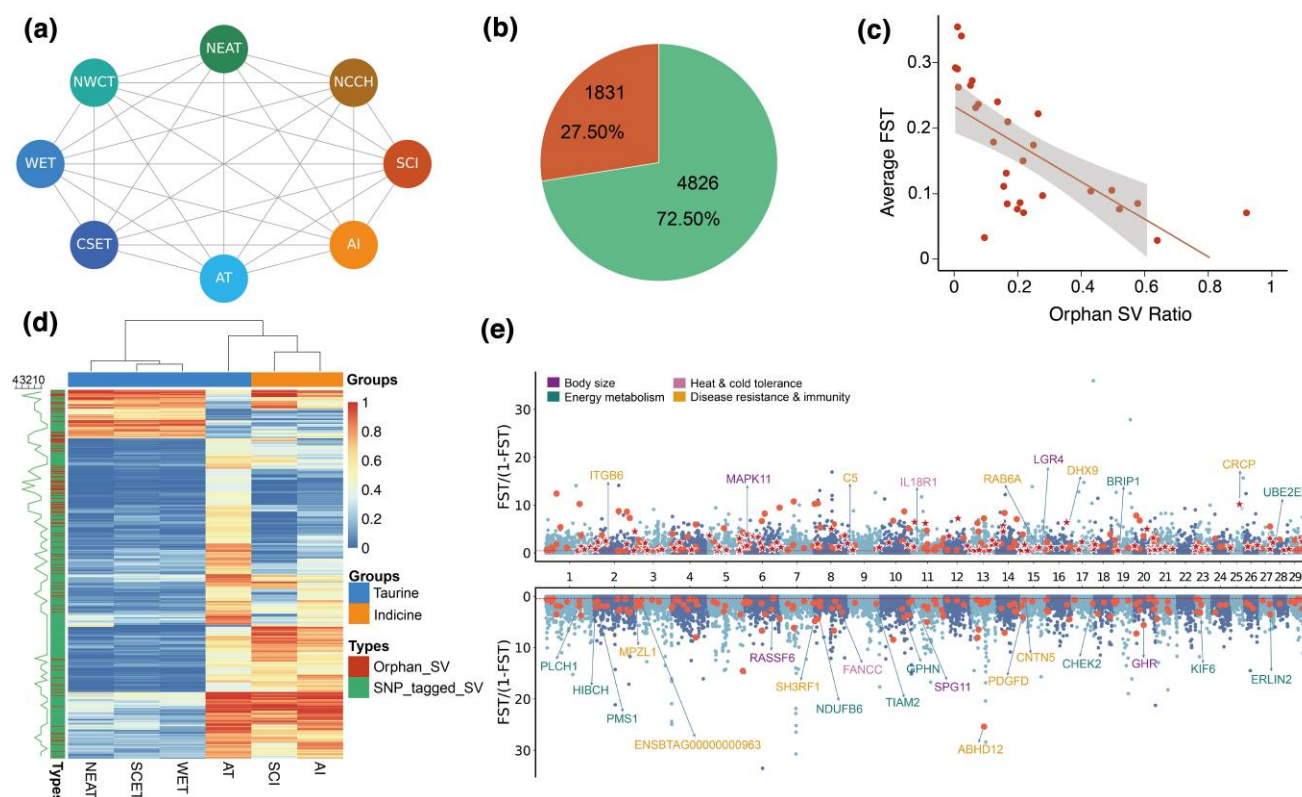


Fig. 5. Selection of orphan SVs among cattle populations from different geographic regions. a) Overview of population comparisons from different regions. b) Pie chart showing proportions of SNP-tagged SVs and orphan SVs. c) Correlation analysis between average F_{ST} value and orphan SV ratio across population comparisons. d) Heatmap of significantly selected SVs between African and other taurine populations. e) Manhattan plot for the selection signatures of SVs in African taurine vs. European taurine and African indicine vs. European taurine. SVs with $F_{ST} > 0.3$ in both of the two comparisons were marked as red dots. SVs that only showed an $F_{ST} > 0.3$ between African taurine and European taurine were marked as red stars. Functions of genes were annotated using DAVID online software. The genes related to various adaptabilities were marked using different colors.

selected between two different populations (Fig. 5a and supplementary table S16, Supplementary Material online). Among these, 1,831 SVs (27.50%) were classified as orphan SVs (Fig. 5b). Interestingly, we observed that the proportion of orphan SVs showed a negative correlation with the genetic divergence (average F_{ST} value) between the populations compared (Fig. 5c). Compared to SNP-tagged SVs, orphan SVs displayed significantly lower F_{ST} values and MAF (supplementary fig. S9a and b, Supplementary Material online) and exhibited patterns according to geographic locations rather than subspecies formation (supplementary fig. S9c and d, Supplementary Material online). This again suggest that the orphan SVs might represent recent mutations that have spread among cattle herds in the local surrounding area due to their adaptability to selection.

African taurine, thriving in hot, humid environments, may be due to convergent evolution and benefits from hybridization with indicine (Kim et al. 2020, 2023). Among 926 commonly strong ($F_{ST} > 0.3$) SV selection signatures between African and other taurine, many were with similar frequency between African taurine and indicine but not with other taurine. These SVs might partially explain the adaption of African taurine to a locality like African indicine (Fig. 5d). There were 19 genes related to heat/cold resistance, disease resistance/immunity, energy metabolism, and body size that overlapped with commonly selected SVs for African taurine and African indicine with other taurine (Fig. 5e). It's noted that the orphan SVs among the selective signatures commonly showed medium frequency in the African taurine but with low/no

frequency in other cattle populations (Fig. 5d). Moreover, those SVs were overlapped with genes for heat/cold resistance, disease resistance/immunity, etc. (Fig. 5e). For example, an orphan SV overlapping *CRCP* (chr25:28034437–28034687), a key immunity-related gene, was present in 41.66% of African taurine and a small proportion of African indicine possibly because of local hybridization (supplementary fig. S11, Supplementary Material online). Those orphan SVs might be originally from African taurine, which would be a benefit to help African taurine to adapt the local environment.

Dominant Role of SV in Regulating IGFBP7+ Adipocyte Formation to Coordinate European Cattle Differences and World Cattle Adaptability

A majority of SVs in our study were tagged by SNPs. However, SVs may play more dominant regulatory roles due to their longer sequences, which provide greater potential as regulatory elements compared to SNPs. Using the same thresholds, certain SVs but not the surrounding SNPs could be detected, and these selection signals would likely be missed in traditional SNP-based analyses. For further investigation, we selected an SV (chr6:72419812–72422015) that was significantly selected in European improved cattle and across various regional cattle populations. This SV exhibited a high presence frequency in MS beef cattle compared to cattle populations with lower fat deposition ability (large beef cattle and dairy cattle). Its presence frequency decreased progressively as the cattle populations moved from colder to hotter regions

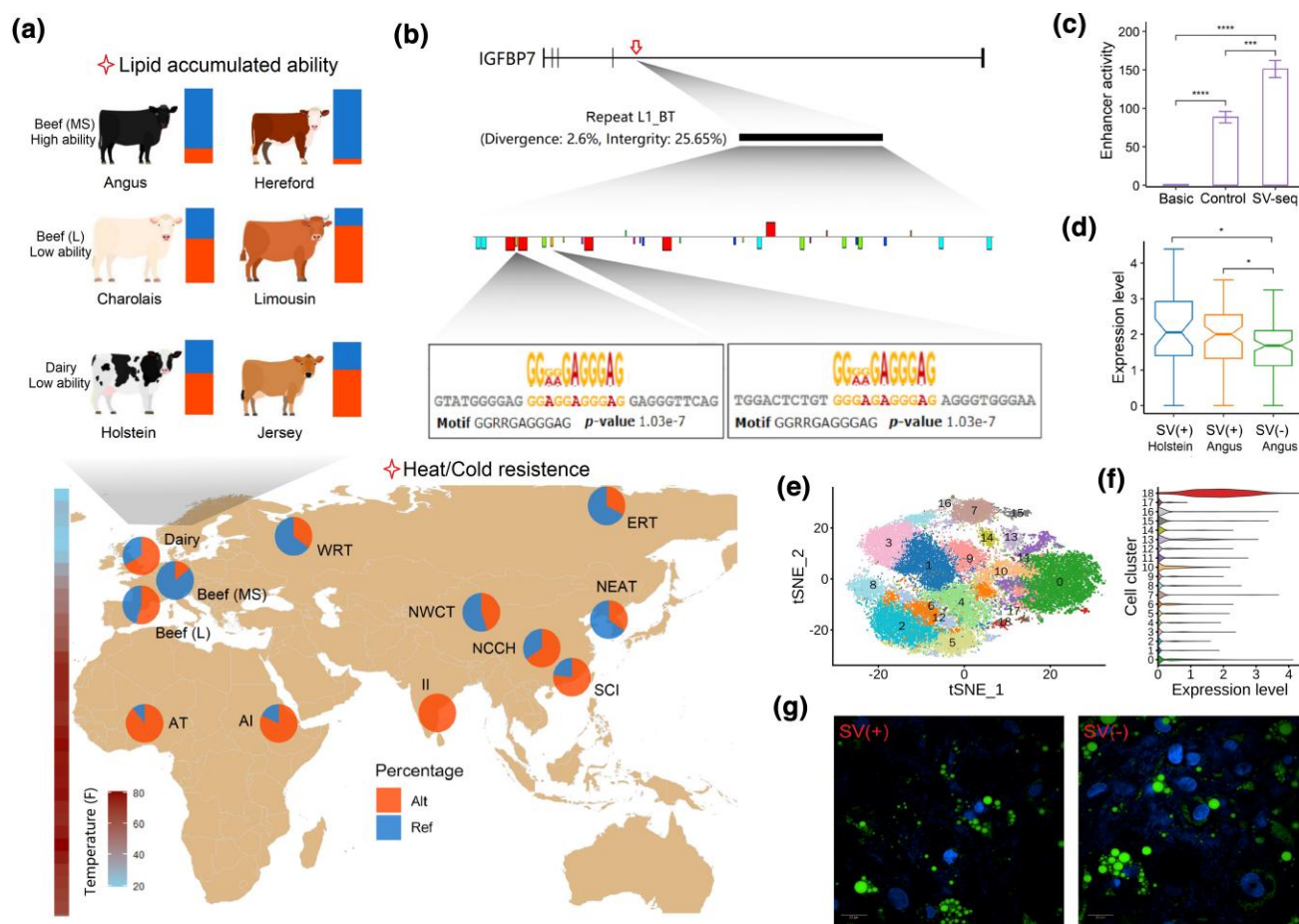


Fig. 6. Functional analysis of an SV in *IGFBP7* and its potential regulation mechanisms for fat deposition. a) Frequency distribution of the SV across cattle populations from different regions and among European improved beef and dairy cattle. Dairy: European improved dairy cattle; Beef (MS): European improved medium-small beef cattle; Beef (L): European improved large beef cattle; WRT, West Russian taurine; ERT, East Russian taurine; NAT, Northeast Asian taurine; NWCT, Northwest Chinese taurine; NCCH, North-Central Chinese hybrid; SCL, South China indicine; II, Indian indicine; AT, African taurine; AI, African indicine. The average temperature for each geographical region was shown on the left side of the map. b) Location and sequence characteristics of the SV. c) Dual-luciferase reporter assay for enhancer activity of the SV sequence. The student's t-test was used for the significance test. *** $P < 0.05$; **** $P < 0.001$; ***** $P < 0.0001$. d) Expression levels of *IGFBP7* in Cluster 18 cells from cattle with different SV genotypes. e) t-SNE analysis of cell clusters. f) Expression levels of *IGFBP7* across cell clusters. g) Comparison of lipid accumulation in primary adipocytes from the SV(-) and SV(+) cattle adipose tissues at Day 12 after the induction for adipogenesis.

(Fig. 6a). Tagged by SNPs, this SV demonstrated the strongest selection signals relative to surrounding SNPs (supplementary fig. S12, Supplementary Material online). At the same time, SNPs near this SV displayed haplotype variations among different cattle populations, providing further evidence for the significance of this selective genomic region (supplementary fig. S13, Supplementary Material online). This SV is located in the first intron of *IGFBP7*, a gene with a critical role in lipid accumulation (Fig. 6b). Its selective characteristics were consistent with energy metabolism traits in MS beef cattle and their adaptability to cold and heat resistance. We validated the existence of the SV using PCR and the IGV software (supplementary fig. S14, Supplementary Material online). The SV sequence was identified as a partial L1_BT element with high divergence (Fig. 6b). Scanning the SV sequence using MEME software revealed multiple motifs, including the GRRRGAGGGAG motif, which was significantly enriched with transcription factors involved in glucose and lipid metabolism, apart from growth factor activity (Fig. 6b and supplementary fig. S15, Supplementary Material online). To understand its functional role, we isolated primary adipocytes from cattle with different SV genotypes and performed

single-cell RNA sequencing. We observed a small cluster of cells (Cluster 18) that uniquely and highly expressed *IGFBP7* (Fig. 6e and f). By applying a dual-luciferase reporter assay, we confirmed that this SV sequence exhibited significant enhancer activity (Fig. 6c). Furthermore, the *IGFBP7* expression was significantly lower in Cluster 18 cells from SV(-) cattle compared to those from SV(+) cattle (Fig. 6d and supplementary fig. S16, Supplementary Material online). This suggests that the SV functions as an enhancer regulating *IGFBP7* expression in Cluster 18 cells.

IGFBP7 is known to be a secreted protein and regulates adjacent cells by binding to receptors in the cell membrane. Specifically, *IGFBP7* binds to the IGF-1 receptor and blocks its activation by insulin-like growth factors (Evdokimova et al. 2012). Differentially expressed genes between *IGFBP7* + cells and other cell types were significantly enriched in pathways related to biological regulation, regulation of cellular processes, response to stimuli, and cell communication (supplementary fig. S17, Supplementary Material online). Previous studies have also shown that certain stromal cell populations can regulate adipogenesis (Schwalie et al. 2018), supporting the hypothesis that *IGFBP7* + cells may act as

regulatory cells. Inducing adipogenesis analysis revealed that primary adipocytes separated from SV(−) cattle adipose tissue had higher lipid accumulation ability than primary adipocytes from SV(+) cattle (Fig. 6g and [supplementary fig. S18, Supplementary Material](#) online). Therefore, we hypothesize that this SV functions as an enhancer to upregulate *IGFBP7* expression in Cluster 18 cells, resulting in elevated secretion of IGFBP7 into the extracellular matrix. This secreted IGFBP7 may modulate lipid accumulation by interacting with IGF1R receptors, potentially through effects on glucose uptake ([supplementary fig. S19, Supplementary Material](#) online).

Discussion

Cattle are central to agriculture and have evolved under natural environmental pressures and human-driven selective breeding ([Rexroad et al. 2019](#)). Selective forces (natural and human-imposed) and nonselective forces (demographic events and introgression) have shaped the cattle genome, influencing traits related to survival, productivity, and market demands ([Decker et al. 2009, 2014](#)). This study integrates SNP and SV analyses for a comprehensive understanding of the genetic factors driving cattle diversity and selection. The use of a pangenome approach captures SVs missed by a single genome. As seen in the latest T2T cattle work (<https://www.researchsquare.com/article/rs-6068440/v1>), which added 431 Mb of new sequences, the number of new bases will likely increase as the linear reference improves and more T2T assemblies are included. This highlights the growing importance of pangenomes in capturing genetic diversity.

Population Structure and Genetic Diversity

Principal component and admixture analyses revealed distinct genetic clusters separating taurine and indicine lineages, reflecting their evolutionary divergence and domestication histories ([Decker et al. 2009, 2014](#)). While taurine breeds are adapted to temperate climates and display high productivity in beef and dairy traits ([Hayes et al. 2009](#)), indicine breeds are adapted to tropical climates with resilience to heat, parasites, and diseases ([Cooke et al. 2020a; Cooke et al. 2020b](#)). Admixture patterns between these two subspecies indicate historical crossbreeding, aiming to combine desirable traits from both lineages ([Cooke et al. 2020a](#)).

SVs contributed significantly to genetic diversity. We observed that ~40.14% of SVs were not tagged ($r^2 < 0.6$, $MAF \geq 0.01$) by nearby SNPs within 1 Mb, emphasizing their unique role in phenotypic variation. Certain SVs cannot be tagged by SNPs ([Korn et al. 2008; Handsaker et al. 2015](#)), as shown in grapevine studies where SVs improved trait heritability ([Liu et al. 2024](#)). We further examined the characteristics and potential formation mechanisms of these orphan SVs. They were mostly associated with MEI-related elements, including L1_BT (43.51%), BovB (13.25%), and BovA2 (10.32%), all of which were unique to ruminants. L1_BT exhibited smaller average divergence compared to other MEIs, suggesting that these SVs may represent recent mutations ([Adelson et al. 2009](#)).

An orphan SV in the *LRP1B* gene was only detected in European taurine, supporting the idea that some orphan SVs may be recent and localized. Orphan SVs typically displayed low selective signals with medium MAFs in specific populations. We observed that populations clustered more by geography than taxonomy and speculated that orphan SVs in

African taurine may contribute to local adaptability. If these SVs align with natural or artificial selection, they may undergo positive selection and expand within subpopulations. Thus, some orphan SVs could represent recent, favorable variants selected in specific populations. Further evidence is needed to clarify their formation mechanisms.

Even when SVs are tagged by SNPs, they may still exert a dominant and causal influence on phenotypic traits. For example, we observed that some SVs, such as those in *GHR* and *IGFBP7*, displayed the highest divergence values in the SNP-SV shared selective regions. We conducted an in-depth study to explore the function of the SV in *IGFBP7* and finally showed that this SV may act as an enhancer for *IGFBP7*, regulating lipid accumulation in an IGFBP7+ cell cluster. This discovery illustrates how SVs can reveal previously unrecognized genetic mechanisms influencing economically important traits. At the same time, we need to note that this SV in *IGFBP7* was significantly selected both in European improved cattle and across various regional cattle populations. In European improved cattle, the SV may regulate fat deposition in MS cattle, which are prone to fat deposition, compared to other cattle. However, in southern China and Africa, where hot environments prevail, the SV may modulate energy deposition in response to environmental pressures, as less energy is required for fat deposition compared to colder regions. Therefore, these findings underscore the importance of considering population-specific characteristics when utilizing selected loci for breeding and management strategies.

Selection and Adaptive Genetic Signatures

Selective sweep analyses have identified genomic regions under strong selection in European improved beef and dairy breeds, driven by human breeding priorities aimed at maximizing productivity. For example, previous findings include genes involved in milk yield and quality (e.g. *DGAT1* and *CSN2*) and genes for growth and muscle development (e.g. *MYF5* and *GDF8*) ([Ma et al. 2021; Mohammadabadi et al. 2021](#)). Functional enrichment was conducted to link these regions to key economically important traits such as milk production, carcass weight, and muscle development ([Qanbari et al. 2014](#)).

Selective signature analysis for European and regional cattle populations identified numerous SNP and SV loci related to economic traits and environmental adaptability. This not only identified numerous SNP loci related to economic traits (milk and beef production) and adaptability among different cattle herds but also uncovered the previously underexplored SV loci. Both selected SNPs and SVs appear to play critical roles in driving variation across cattle populations. This provides further important evidence for the crucial role of SV in explaining the diversity of cattle and valuable genomic resources for a balanced enhancement of the economic value of cattle worldwide in the future.

We also identified untagged SVs by SNPs under significant selection across different cattle populations. The annotated genes and QTLs overlapping with these SVs could explain the phenotype differences in ways similar to SNPs. This indicates that SV plays an indispensable role in understanding cattle diversity, particularly in areas where SNP signals alone are not enough ([Xu et al. 2014; Chen et al. 2024](#)). Our results revealed the potential influence of introgression in shaping SV landscapes and supplied vital information to promote the understanding of adaptation and phenotype differences

between taurine and indicine cattle at the SV level. Additionally, our study provided proof of concept for utilizing SVs as important markers in evolutionary studies and breeding programs aimed at enhancing the adaptive potential of local cattle populations.

Regional-specific analyses further identified adaptive genetic signatures linked to environmental resilience, consistent with previous findings in African and South Chinese cattle (Gao et al. 2017; Kim et al. 2017; Liu et al. 2020; Tijjani et al. 2022; Ayalew et al. 2023). For example, genes like *HSP70*, associated with heat resistance, help support thermoregulation in high-temperature environments (Basiricò et al. 2011). In African cattle, immune-related genes provide resistance to endemic diseases such as trypanosomiasis (Rajavel et al. 2020). These findings provide essential insights into breeding strategies for climate resilience and disease resistance. However, adaptation often comes with tradeoffs. For instance, the enhanced thermotolerance of indicine breeds can lead to reduced milk production compared to taurine breeds (Cooke et al. 2020a). In heat-stressed Holstein cows, three genomic SNPs in the genes *TLR4*, *GRM8*, and *SMAD3* have been identified as molecular markers for both milk production and thermotolerance (Zamorano-Algandar et al. 2023). Understanding these tradeoffs is essential for designing breeding programs that balance productivity with resilience, ensuring sustainable improvements in cattle genetics for diverse environments.

Implications and Applications

This study demonstrates the independent and complementary roles of SVs vs. SNPs in understanding genetic diversity and selection. While SNPs are crucial for detecting single-base changes and associated traits, SVs capture the large variations that often escape SNP-based analyses. For instance, the *IGFBP7* SV highlights how structural variations can influence phenotypic traits independently of SNP signals. The findings have significant implications for cattle breeding and improvement. Incorporating SVs into genomic selection pipelines will enhance the accuracy of breeding programs, particularly for traits where SNP signals are insufficient (Hayes et al. 2009). Additionally, insights into adaptive genetic signatures can guide breeding strategies to improve resilience against climate change and disease pressures, which is especially relevant for sustainability in developing regions (Akinsola et al. 2024). Developing countries face unique challenges in cattle production, including limited resources, harsh environments, and diverse disease pressures. Identifying regional-specific genetic adaptations provides opportunities to develop breeds tailored to these conditions. Enhancing heat and disease resistance through targeted breeding or gene editing offers a promising pathway to boost productivity and food security (Liu et al. 2022).

Limitations and Future Directions

The improvements in SV calling accuracy and resolution in this study remain constrained by limitations such as reliance on short-read sequencing and low coverage in samples. Short reads are particularly ineffective for detecting large duplications and inversions, and mapping challenges persist in repetitive or complex genomic regions. Long-read sequencing technologies, like PacBio or Oxford Nanopore, offer significantly higher sensitivity for structural variant detection, with studies showing up to fivefold improvements compared to

short-read methods (Huddleston et al. 2017; Miga et al. 2020; Cheng et al. 2021; Ebert et al. 2021; Logsdon et al. 2021; Bai et al. 2022; Olagunju et al. 2024; Su et al. 2024). Future efforts should prioritize improving reference genome quality, incorporating high-quality assemblies into graph-based pangenomes (Nguyen et al. 2023; Smith et al. 2023; Kalbfleisch et al. 2024; Zhang et al. 2024), and expanding datasets to include underrepresented cattle populations, especially indigenous breeds. Cattle graph pangenomes, which already show advantages in structural variant detection, hold great promise for enhancing genome-wide studies (Leonard et al. 2022, 2024). Additionally, experimental validation of candidate SNPs and SVs, through tools like CRISPR-based editing, will be essential for confirming functional roles. As sequencing costs decrease, integrating long-read data and other structural variants such as inversions and translocations will provide a more comprehensive understanding of cattle genetic diversity, adaptation, and evolutionary processes.

Conclusion

Integrating SNP and SV analyses provides a powerful framework for unraveling the complex genetic architecture of cattle populations. This approach identifies distinct genetic clusters for taurine and indicine lineages, genomic regions under selection for economically important traits such as milk production and muscle development, and unique contributions of SVs, including regulatory roles in fat deposition and other traits. The enhanced SV catalog generated here serves as a vital resource across diverse cattle breeds, shedding light on genomic content changes during domestication, breeding, and improvement. These insights highlight the potential of integrating SNP and SV analyses to unravel complex genetic networks driving cattle diversity and selection. This approach offers transformative opportunities for breeding programs to tackle food security challenges in developing countries and foster sustainable livestock management in the face of climate change.

Materials and Methods

Whole Genome Sequencing Data Collection and SNP Calling Processes

The 2,409 cattle of 82 cattle breeds were retrieved from the NCBI Sequence Read Archive (SRA; <https://www.ncbi.nlm.nih.gov/sra/>) and our previous studies (Zhou et al. 2022). The accession ID for each data set is in [supplementary table S1, Supplementary Material](#) online. Adapters and low-quality reads were filtered using parameters of “-q 20 -u 30 -l 75” by *fastp* (V0.23.2) (Chen et al. 2018b). The cleaned reads were then aligned to the cattle reference genome (ARS-UCD1.2) and SNPs were called using the BWA-GATK pipeline in Sentieon tools (<https://www.sentieon.com/>) (V202308). For the initial quality control of variants, VCFtools (V0.1.16) (Danecek et al. 2011) and GATK (V4.3.0) (Van der Auwera et al. 2013) were used with filtration criteria for SNPs: (i) max-missing 0.3; (ii) max-alleles 2; (iii) maf 0.01; (iv) min-meanDP 3; (v) QD < 2.0; (vi) QUAL < 30.0; (vii) SOR > 3.0; (viii) FS > 60.0; (ix) MQ < 40.0; (x) MQRankSum < -12.5; and (xi) ReadPosRankSum < -8.0.

Pangenome Graph Construction and SV Calling

To get more SVs genotyped for collected samples, we employed a strategy combining SVs called by different assemblies

and high-quality SVs detected by short reads in large populations. First, we identified insertions and deletions for 23 collected cattle genome assemblies by pairwise comparing them to the reference genome (ARS-UCD1.2) through three assembly-based SV calling tools: (i) Svrifine (V0.35), with the parameter “--maxsize 1,000,000”; (ii) Assemblytics (V1.2.1), with the “unique_anchor_length” set to 10,000, “min_variant_size” set to 50, and “max_variant_size” set to 10,000; and (iii) Minimapp2 (V2.25) (Li 2021), used to map the assembly to the reference with the parameter “asm5”. The output PAF file was used to call SV using *paftools*, with the parameter “-L 10,000.” To improve SV breakpoint accuracy, we realigned sequences 100 bp upstream and downstream of the SV boundaries detected by the three software to the local reference using both *mafft* (V7.487) (Katoh and Toh 2010) and *msa2vcf* (V1.0). Second, we incorporated 92,518 high-quality deletions identified through WGS in 898 cattle samples across 57 different breeds in our previous study (Zhou et al. 2022). Finally, we used the ARS-UCD1.2 assembly as the backbone of the pangenome graph. All merged SVs were incorporated into a variant graph using the “construct” module of the *vg* toolkit (V1.51.0) without removing any alternate alleles (Hickey et al. 2020). The resulting pangenome graph was then indexed in XG and GCSA formats using “vg index”, with the “-L” parameter enabled for both formats. SVs were depicted as bubbles in the graph, with paths representing the corresponding alleles.

Each sample's clean reads were mapped against this graph genome using *vg Giraffe*, resulting in alignments in the GAM format (Sirén et al. 2021). Alignments with mapping quality below five or base quality below five were filtered. A compressed coverage index was calculated using “vg pack”, and snarls were created using “vg snarls,” both with default parameters. Finally, SV genotyping results for all samples were produced using “vg call,” with the parameter “-v --bias-mode --het-bias 2,4” on the constructed pangenome graph.

Population Structure Analysis

We prepared data for population analyses by randomly selecting 50 individuals for breeds with sample numbers over 50 and retaining breeds with sample numbers less than 50. Mutations with MAF < 0.01 or animals with low call rates (< 0.8) were filtered out. We then performed LD-based pruning genotype data by filtering mutations with $r^2 > 0.5$ using PLINK. This resulted in data for 472 samples from 14 cattle populations for 1,667,536 SNPs and 86,775 SVs. PCA was performed using “-pca” option of the PLINK software (v1.9) (Purcell et al. 2007). ADMIXTURE was run for each possible group ($K=2$ to 10) with 200 bootstraps using Admixture (v1.3.0) (Alexander et al. 2009). The neighbor-joining tree was constructed using the matrix of pairwise genetic distances calculated by *VCF2Dis* (v1.4.0). A population-level phylogeny was reconstructed using the maximum likelihood (ML) mode in TreeMix (v1.13).

Genetic Diversity Evaluation, LD Analysis, and ROH Detection

Genome-wide nucleotide diversity and genetic distances (F_{ST}) of different cattle populations were estimated using VCFtools (v0.1.17). The information of population and variation used for selective signatures was listed in [supplementary table S5, Supplementary Material](#) online. LD decay was calculated

using *popldelay*. To ensure comparability between European improved cattle and other European cattle, pi-hat values were calculated using PLINK, and animals with pi-hat values higher than 0.5 were filtered out. Population sizes were balanced by retaining a similar number of animals. ROH was identified using Plink1.9 with “-homozyg” option and default parameters: “-homozyg-window-snp 20 -homozyg-density 50 -homozyg-window-het 1 -homozyg-window-missing 0 -homozyg-snp 20 -homozyg-kb 100 -homozyg-gap 50”. ROH islands were defined as genomic regions where at least 80% of the animals in a population had consecutive SNPs in an ROH (Pemberton et al. 2012).

Selective Signature Analysis for SNPs and SVs

We used a strategy by combining single-site-based and multiple window-based selective signature identification methods to minimize false positives. First, F_{ST} values were calculated for each SNP between compared cattle populations using VCFtools (v0.1.17). Then, three methods (F_{ST} , XP-EHH, and XP-CLR) were applied to detect selected genomic regions using SNPs. F_{ST} , average normalized XP-EHH scores, and XP-CLR were calculated for 50-kb windows with 20-kb steps using VCFtools (v0.1.17), selscan (v1.3.0), and xpclr (v1.1.2), separately. Only windows shared in the top 1% across the two methods were considered putative significant windows. High-confidence selected windows were defined as those containing at least one SNP in the top 1% of F_{ST} values. For the SVs, single-site F_{ST} calculations were conducted using VCFtools (v0.1.17). The top 1% was recognized as significant SVs. When comparing taurine and indicine cattle, only cattle with over 90% pure lineage were included.

Gene Function and QTL Enrichment

The gene annotation file was downloaded from the Ensembl database (https://ftp.ensembl.org/pub/release-80/gtf/bos_taurus/). Cattle QTL information was downloaded from the Animal QTL database (<https://www.animalgenome.org/cgi-bin/QTLdb/index>). Gene functional annotation analyses were performed using the online DAVID software (<https://david.ncifcrf.gov/>). Fisher's exact test was conducted to measure gene enrichment in annotation terms (threshold: 0.05). QTL enrichment analysis in particular regions (selected SVs) was conducted by Genome Association Tester software with 1,000 simulations of randomly assigned genomic intervals of the same size (Hege et al. 2013). Enrichments with FDR < 0.05 were considered statistically significant.

Presence/Absence Analysis for the Selected SV Sequence Located in *LRP1B*

We downloaded 16 representative assemblies for ruminant suborder, which included GCA_014182915.2 for Gaur, GWHBDOC00000000 for Gayal, GCF_032452875.1 for Banteng, GCF_002263795.3 for Cattle, GCF_003369695.1 for Zebu Cattle, GCA_005887515.3 for Domestic Yak, GCF_000754665.1 for American Bison, GCA_963879515.1 for European Bison, GCF_019923935.1 for Water Buffalo, GCF_001704415.2 for Goat, GCF_016772045.2 for Sheep, GCA_017591445.1 for Giraffe, GCF_910594005.1 for Red Deer, GCF_022376915.1 for Chinese Forest Musk Deer, GCA_022376925.1 for Lesser Mouse Deer, GCF_949987535.1 for Minke Whale. We constructed a phylogenetic tree of species genetic distance for these assemblies based on

the Mash algorithm. Then, Cactus (v2.9.0) was used to perform whole genome alignment for all species to obtain the .maf file (Hickey et al. 2024). We checked for the sequence of the SV in each species to determine its presence or absence.

Targeted-PCR Validation of the L1_BT Insertion in *IGFBP7*

Genomic DNA for one Holstein and two Angus cattle was extracted from preadipocytes following the instructions of the DNA extraction kit (Tiangen, Beijing). Primers were designed (forward: GCCAGGGGTCTTAGTCT; reverse: AGGTTCCGATCCACGATA) based on the cattle genome version (ARS-UCD1.2). PCR amplification was performed with a 50 mL reaction volume according to the Taq DNA polymerase manufacturer's protocol (2×Hieff PCR Master Mix, Yeasen, Shanghai). The genomic DNA was amplified on a Bio-Rad MyIQ thermocycler. The touchdown PCR cycle for target region amplification was as follows: initial denaturation at 94 °C for 4 min; followed by 18 cycles of 94 °C for 30 s, annealing at 68 °C ~ 50 °C (decrease 1 °C per cycle) for 30 s; 22 cycles of 94 °C for 30 s, annealing at 50 °C for 30 s; primer extension at 72 °C for 1 min; and final extension at 72 °C for 10 min. All the amplified products were run on a 1.5% agarose gel.

Single-cell RNA Sequencing for Cattle Primary Adipocytes

Cattle primary adipocytes were isolated from subcutaneous adipose tissue following our previously established protocols (Peng et al. 2023). The subcutaneous adipose tissues were collected under the approval of the Animal Experimental Ethical Inspection of Laboratory Animal Center, Huazhong Agriculture University with the ID number HZAUCA-2022-0010. The cells were cultured in high-glucose DMEM medium (4.5 g/mL Glucose, 4.0 mM L-glutamine, Cytiva) containing 10% FBS supplemented with 1% penicillin-streptomycin at 37 °C and 5% CO₂. When the density of the cattle primary adipocytes reached 70%, cells were collected and washed with resuspension buffer. Cells were filtered through a 40 µm strainer to remove cell clumps. Cell viability was determined to be 90% using trypan blue staining and hemocytometer counting. Three libraries were constructed for cattle primary adipocytes isolated from Holstein cattle PAV(+), Angus cattle PAV(+), and Angus cattle PAV(–) using the Chromium Single Cell 3' Gel Bead-in-Emulsion, Library & Gel Bead Kit v3 (10× Genomics, Pleasanton, CA, United States). Each scRNA-seq library was paired-end sequenced in a single cell flow lane on an Illumina HiSeq system.

The Cell Ranger (v7.1.0, 10× Genomics) single-cell software was used to perform sample demultiplexing, alignment, filtering, and UMI counting. Cells are excluded based on the 5 median absolute deviation from the median value of detected genes across all cells and more than 10% of total UMIs mapping to the mitochondrial genome. Potential cell doublets were removed using DoubletFinder (v2.0) with default parameters and 7.5% homotypic doublet proportion estimation (McGinnis et al. 2019). Data analysis was performed using the R package Seurat (v5.1.0) in the R environment (v 4.1.1), including quality control, normalization, and scaling of data, feature selection, dimensionality reduction, clustering, and visualization of data (Hao et al. 2021). In Seurat, data were normalized using the “NormalizeData,” 2000 highly variable genes were identified using “FindVariableFeatures” and data were scaled using “ScaleData.” The remaining highly variable genes were

decomposed by PCA, and the top 50 dimensions were selected. The 50 PCs were harmonized across samples using Harmony (v0.1) (Korsunsky et al. 2019). Cells were clustered into subpopulations according to the same dimensions using the “FindClusters” function with a 1.0 resolution.

Function Prediction and Enhancer Activity Analysis for the L1_BT Insertion Sequence in *IGFBP7*

The MEME online tool (<https://meme-suite.org/meme/>) was used to search for motifs in the sequence of L1_BT in the *IGFBP7* gene. The top 10 motifs were identified and submitted to the GOMo module to identify possible roles (Gene Ontology terms) for each motif. Enhancer activity of the L1_BT insertion sequence in *IGFBP7* was analyzed using a luciferase reporter assay in HEK293T cells. Briefly, cells were seeded in a 96-well plate. When the cell confluence reached ~70% to 80%, the vectors were transfected into the cells using transfection reagents (PEI MW 25000, Shanghai). In the luciferase reporter system, the vector pRL-TK served as an internal reference, while the pGL3-Control and the empty pGL3-Basic were used as positive and negative controls respectively. The activities of firefly luciferase and Renilla luciferase were measured according to the manufacturer's instructions for the luciferase reporter assay kit (Yeasen, Shanghai). After measuring the two fluorescence values, the ratio of the fluorescence intensity of firefly luciferase to that of Renilla luciferase was calculated. The final data of the experiment were obtained and presented in the form of mean ± standard error.

Lipid Accumulation Ability Comparison Between Cattle With Presence/Absence Status of the L1_BT Insertion Sequence in *IGFBP7*

Cattle primary adipocytes, with six replicates from Angus cattle with different SV genotypes, were cultured in a high-glucose DMEM medium containing 10% FBS and supplemented with 1% penicillin-streptomycin at 37 °C and 5% CO₂, separately. The culture medium was changed every 48 h. When cell confluence reached 90%, a differentiation-inducing medium containing 1.0 µmol/L dexamethasone, 0.5 mmol/L 3-isobutyl-1-methylxanthine, 1.0 µmol/L rosiglitazone, and 10 mg/L insulin was applied for 48 h. Subsequently, a fresh medium containing 10 mg/L insulin and 1.0 µmol/L rosiglitazone was used to maintain differentiation. On the 12th day, the cells were fixed in 2% formaldehyde at 4 °C, then washed three times with PBS for 5 min at 4 °C. Fixed cells were stained with 7.5 µM BODIPY and 28.55 µM DAPI for 30 min at room temperature. For each replicate, 45 images were captured using fluorescence microscopy to quantify and compare lipid accumulation in cattle primary adipocytes with different SV genotypes.

Supplementary Material

Supplementary material is available at *Molecular Biology and Evolution* online.

Author Contributions

All authors have read and approved the manuscript. Y.Z., G.E.L., and L.Z.F. conceived and designed the experiments. Y.Z., S.L.D., P.J.Z., W.H.L., L.W.P., X.Z., W.G.Z., X.L.D., and L.J.X. performed in silico prediction and computational analyses. Y.Q.D. performed PCR confirmation. X.Y.L. and E.H.J. performed enhancer activity analysis. Y.Z., S.L.D.,

G.E.L., L.Z.F., P.J.Z., Z.Q.L., L.Y., W.F.L., and L.G.Y. collected samples and generated genome sequencing data. Y.Z., S.L.D., G.E.L., L.Z.F., and P.J.Z. wrote and revised the paper.

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Conflict of Interest

All authors declare no potential conflict of interest.

Data Availability

The 2,409 WGS data of 82 cattle breeds were retrieved from the NCBI database (<https://www.ncbi.nlm.nih.gov/sra/>) and our previous studies. The accession ID for each dataset is in [supplementary table S1, Supplementary Material](#) online. The single-cell RNA sequencing data generated in this study have been submitted to the NCBI (<https://www.ncbi.nlm.nih.gov>) under accession number PRJNA1198857. The pangenome graphs and SNP-SV joint reference panel for the current study are available from Figshare (<https://doi.org/10.6084/m9.figshare.28021577>).

All software and their respective versions used in the study are publicly available as described in the Methods section. The pipelines for pangenome construction, SNP calling, SV genotyping, etc. are available at <https://github.com/PengjuZ/CattleSV>.

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